E2 Results — The Probe's Own Ledger: Balanced, or Net Stored?

Madness in the Probe sub-programme · design experiment · banking run · July 2026 Alexander Pisani & Claude (Anthropic)

Status

ALL 22 COMMITTED CHECKS PASSED. 6 report-only / labelled items reported. Banking run complete.

The pre-registered prediction — *all committed measures pass; the identities are classical and the deliverable is the resolution plus the three priced quantities* — held in full. No committed bar was moved; no result was patched. The originating question is answered.

The resolution. *Balanced, or net stored?* resolves as a **conservation frontier**. On the probe's own ledger the ingested information of a stream is split, with no residue, between what the probe **pays to erase** and what it **retains in storage** — and the split is the free parameter. There is no regime in which the ledger fails to close and no regime in which it stores "net energy" beyond the information it took in: storage is exactly memory growth, payment is exactly memory bound, and the two poles are the **summarizer** (erase everything, pay the full rate, store nothing) and the **hoarder** (keep everything reversibly, pay nothing, store the full rate). What knowledge changes is not *whether* it balances but the *rate*: a learned model is a discount on the price of forgetting worth exactly the redundancy it captures, and a wrong model is a surcharge of exactly the KL divergence.

Implemented fresh for E2 only. No prior script was consulted or reused; no other corpus document was read. The pre-run pins C1–C4 (brief §6) are honoured exactly. Implementation choices within the implementing instance's remit are documented in §Reproducibility and §Notes.

Reproducibility block

Item	Value
Script	`521_e2_ledger.py` (fresh; PID-prefixed)
Recorded seed	`20260710`
Runtime PID (this run)	`565`
Wall-clock runtime	~1.4 s
Exit code	`0` (all committed checks passed)
Python	3.12.3
numpy	2.4.4
mpmath	1.3.0

Item	Value
Precision	float64 `math.lgamma` + `math.fsum` throughout; one **mpmath dps=50** cross-check of the M1 identity at n=100
Stream length `T`	100,000 (i.i.d. from `p`, one seeded stream shared by M1-realized / M2 / M3 / M4)
Markov stream (M6)	100,000 from a separate seed (`SEED+1`)
M2 memory bound	16 count records (end-of-run inventory reported as an explicit storage line)

Precision justification (why float64 suffices here). Every quantity is an entropy or a log-count in nats, magnitude ~10 (largest is `n·H` at n=6400 7763). float64 carries ~15–16 significant digits, so absolute error is 10 11 even at the largest scale; `fsum` removes accumulation error. The tightest bar (M1 at 10) therefore clears by 4 orders. This is confirmed independently: recomputing the M1 identity at n=100 in mpmath (dps=50) gives residual **−1.09×10** , i.e. the float64 residual (~10 12) is an honest float artifact, not a masked violation. Charges use the same `lgamma`-based coefficient wherever `op-F` and `op-D` must cancel, so the route-equality (A1) closes to machine epsilon.

Pre-registered bars are copied verbatim into the script header (before any check was written) and are reproduced with actuals below.

Implementation choices (documented, within remit / ratified in §6)

- **Learned model = pinned dyadic `p`** everywhere (C2); tolls are averages of `-ln p(x)` over the realized stream, never an empirical re-fit.
- **op-D on a stored count record** charges `-ln P_multinomial(c)` under `p` (C1) — the self-information of what is still held.
- **M4** is op-R (reversible, free) + archive-or-discard of whole blocks; **op-F does not enter M4** (C3). Archiving = first fB blocks (representative for i.i.d.).
- **M2** memory bound = 16 records; the 16 records still held at the end are reported as an explicit **retained-storage line** so the conservation statement closes (C4, lead's addition).
- **H(counts)** is computed by an independent chain-rule route (binomial entropies of partial sums, $O(n^2)$), **not** by subtracting `E[ln C]` from `nH` (anti-tautology, §5.3), and is validated against brute-force simplex enumeration at n=50 and n=100.
- **Three labelled additions**, graded separately — A1 route-equality, the enumeration validation, and the capacity/blank harvest — see §Implementer additions.

Pre-registered bars vs actuals

Check	Bar	Actual	Verdict
M1 conservation n=50	|nH − (H(counts) + E[ln C])| 1e-9	resid **+1.82e-12**	**PASS**
M1 conservation n=100		resid **−3.82e-12**	**PASS**

Check	Bar	Actual	Verdict
M1 conservation n=200		resid **−3.00e-11**	**PASS**
M1 realized n=50	|mean op-F − E[ln C]|/SE 3	z = **0.86** (B=2000)	**PASS**
M1 realized n=100		z = **0.93** (B=1000)	**PASS**
M1 realized n=200		z = **0.91** (B=500)	**PASS**
M2 perpetual toll	paid erasure/sym H ± 0.02	**1.21317** (H=1.21301, |d|=1.7e-4)	**PASS**
M3 ignorant → ln4	ln4 ± 0.02	**1.38629** (exact, see note)	**PASS**
M3 learned → H	H ± 0.02	**1.21441** (|d|=1.4e-3)	**PASS**
M3 mismatched → H+KL	H+KL ± 0.02	**1.64699** (|d|=7.7e-4)	**PASS**
M3 ordering	learned < ignorant < mismatched	1.2144 < 1.3863 < 1.6470	**PASS**
M4 frontier f=0	storage+erasure = H ± 0.02	**1.21441**	**PASS**
M4 frontier f=¼,½,¾,1		**1.21441** at every f	**PASS**
M4 storage f=0	storage = f · H ± 0.02	**0.00000**	**PASS**
M4 storage f=¼		**0.30459** (f · H=0.30325)	**PASS**
M4 storage f=½		**0.60716** (f · H=0.60650)	**PASS**
M4 storage f=¾		**0.91021** (f · H=0.90976)	**PASS**
M4 storage f=1		**1.21441** (f · H=1.21301)	**PASS**
M5 MDL slope	slope = −1.5 ± 0.05	**−1.5038** (intercept −1.108)	**PASS**

Report-only / labelled additions: M1 chain-vs-enumeration validation; M1 mpmath cross-check; A1 route-equality; M4 capacity/blank harvest; M6 order-0-vs-order-1 gap. (Below.)

Per-measure detail

M1 — dual-accounting conservation (the non-circularity core)

The identity `nH = H(counts) + E[ln C(n;c)]` is the chain rule `H(X ) = H(counts) + H(X |counts)`, with the conditional equal to `E[ln (multinomial coefficient)]` because, given counts, the block is uniform over its class (exchangeability). All three terms are computed by **genuinely different routes**: `nH` from the closed form; `E[ln C]` from per-coordinate binomial sums (`ln n! − Σ E[ln c !]`, each `c ~ Binom(n, p )`); `H(counts)` from a chain-rule over binomial entropies of partial sums. None is obtained by subtracting the others. The identity held to $\mathbf{1.8\times10^{12}}$ (n=50), $\mathbf{3.8\times10^{12}}$ (n=100), $\mathbf{3.0\times10^{11}}$ (n=200) — four to seven orders inside the 10 bar. The chain-rule `H(counts)` matched brute-force simplex enumeration to $\mathbf{1.4\times10^{\ 1}}$ (n=50) and $\mathbf{6.4\times10^{\ 1}}$ (n=100), and the two independent `E[ln C]` routes agreed to ~$\mathbf{4\times10^{\ 12}}$. The mpmath (dps=50) recomputation put the identity residual at **10** . On the realized leg, the mean per-block op-F charge over the T-stream matched the exact `E[ln C]` at **z = 0.86 / 0.93 / 0.91** — comfortably inside 3 SE at all three block sizes.

M2 — the perpetual toll (bounded probe)

The bounded probe forgets order within n=100 blocks (op-F, paying `ln C`) and discards count records under the learned model when memory exceeds 16 records (op-D, paying `-ln P(c)`). The **paid** erasure came to **1.21317 nats/symbol**, within **1.7×10** of H — inside the ±0.02 bar. The full ledger closes exactly:

> paid erasure (1.21317) + retained storage (0.00123) = **ingested rate 1.21441** = the direct per-symbol op-D(raw) codelength to $\mathbf{7\times10^{\ 1}}$.

The 16 records still held at the end carry the 0.00123 nats/symbol of retained storage; the paid toll sits below the ingested rate by exactly that inventory. The ingested rate is H up to Monte Carlo (here $+1.4\times10^{\ 3}$ above H on this stream), which is why the paid toll lands a hair *above* H rather than below — the ledger identity is exact, the relation to the theoretical floor carries the stream's fluctuation. The universal floor (no lossless-forgetting policy pays less than H/symbol in expectation) is **Shannon's**, cited as classical; this run measures a policy's realized toll against it, and does **not** present the measurement as a proof of the floor.

M3 — the knowledge discount and the error surcharge (the experiment's teeth)

Three erasers, op-D on raw blocks of the same realized stream:

- **ignorant** (uniform protocol) → **1.38629** = ln4 (the cell capacity);
- **learned** (protocol p) → **1.21441** H;
- **mismatched** (protocol q) → **1.64699** H+KL,

with the strict ordering **learned < ignorant < mismatched** holding. The **discount** of knowing the distribution is `ignorant - learned = 0.17189`, matching the redundancy `ln4 - H = 0.17329`. The **surcharge** of the wrong model is `mismatched - learned = 0.43258`, matching `KL(p||q) = 0.43322`. The teachable lands sharply: the surcharge (**0.4332**) exceeds the redundancy (**0.1733**), so the confident wrong model costs **more than pure ignorance** — `1.647 > 1.386`. Knowing the wrong thing is worse than knowing nothing.

M4 — balanced-or-stored: the frontier

Whole blocks are reversibly compressed (op-R, free) and then either archived (storage) or discarded under the learned model (op-D). Across `f  {0, ¼, ½, ¾, 1}`:

f	storage/sym	erasure/sym	sum	f · H
0.00	0.00000	1.21441	**1.21441**	0.00000
0.25	0.30459	0.90982	**1.21441**	0.30325
0.50	0.60716	0.60725	**1.21441**	0.60650
0.75	0.91021	0.30419	**1.21441**	0.90976
1.00	1.21441	0.00000	**1.21441**	1.21301

The frontier `storage + erasure` is the ingested rate at **every** f — conservation is f-independent, which *is* the content: no matter how the probe splits keep-vs-discard, the ledger sums to what it took in. Storage tracks `f·H` (within 1.3×10 ^{3}) and the two poles are realized cleanly — summarizer at f=0 (pays H, stores 0) and hoarder at f=1 (stores H, pays 0).

M5 — the model's residue (the MDL term)

The exact `E[ln C(n;c)] - nH` was computed for n from 50 to 6400 and fit against `ln n`: slope **−1.5038**, dead on the pinned `-(|A|-1)/2 = -1.5` and inside ±0.05. The per-doubling decrements converge monotonically to `-1.5·ln2 = -1.0397` (−1.0516 → −1.0451 → −1.0423 → −1.0410 → −1.0404 → −1.0400 → −1.0399), the clean signature of the Rissanen/MDL parametric term. What op-F retains — the counts — is exactly what its charge falls short of `nH` by, and that permanent residue of n symbols grows as `(k-1)/2 · ln n`. What survives forgetting the details is the model, logarithmically.

M6 — Markov extension (report-only, no bar, no grade)

Sticky doubly-stochastic chain (`P_stay=0.7`, else `0.1`; stationary uniform, verified). The **order-0** learned toll (marginal) was **1.38629** = ln4; the **order-1** learned toll (transition) was **0.93641**, estimating the entropy rate `H(0.7,0.1,0.1,0.1) = 0.94045`. The gap **0.44989** reproduces the adjacent-symbol mutual information `ln4 - H(row) = 0.445846` (the ~4×10 ^{3} excess is Monte Carlo on the order-1 estimate). The discount of M3 extends to temporal structure: a higher-order model lowers the toll by exactly the mutual information it captures.

Grades on pass (per §4 of the brief)

- **Fact** (classical) — the conservation identity `nH = H(counts) + E[ln C]` (Shannon's source coding; the chain rule and exchangeability); the toll values ln4, H, H+KL (Shannon's source-coding theorem and the mismatched-code / Kraft identity: erasing p-data under model m costs the cross-entropy `H(p) + KL(p‖m)` per symbol); the KL surcharge; and the MDL slope `-(k-1)/2` (Rissanen). The reversible-free / erasure-positive accounting is Landauer and Bennett. Attributions carried; the thermodynamics-of-prediction literature is adjacent and may be cited as such.
- **Reading** (E2's own, on top of those facts, on the probe's own ledger):
  1. **The resolution** — *"balanced or stored?" is answered by a conservation frontier: storage is exactly memory growth (f · H), payment is exactly memory bound ((1−f) · H), the ledger closes to the ingested information at every f, and the poles are the summarizer and the hoarder.*
  2. **Knowledge is a discount on the price of forgetting**, worth exactly the redundancy the model captures (`ln4 - H`).
  3. **Model error is a surcharge of exactly `KL(p‖q)`** — the probe's model quality is measurable on its own ledger — with the sharp corollary that a confident wrong model costs *more* than ignorance whenever `KL > redundancy`.
  4. **Learning's permanent residue grows as the MDL term** — what op-F keeps (the counts) is the model, and it grows as `(k-1)/2 · ln n`.

Note forward (one line, per §4): M3's "erasure paid per nat" machinery supplies the numerator and denominator vocabulary for the register's **E6** candidate `M(q) = erasure paid / entropy` `removed`.

Implementer additions (labelled; separate grade; do not substitute for any committed measure)

A1 — route-equality (the exact per-block ledger split). For every one of the 1000 blocks in the M2 run, `op-F(c) + op-D(counts)(c) = op-D(raw block)` held to a max deviation of **2.8×10**^{1} (machine epsilon). This is the exact-per-realization identity behind C1: the two forgetting routes — forget-order-then-discard-the-summary versus discard-the-raw-block — cost **identically**, and differ only in *where* the payment is located on the ledger (op-F pays the order information `ln C`; op-D(counts) pays the count information `-ln P(c)`; their sum is the block's self-information `-Σ ln p(x )`). It is the demonstration that M2's accounting is a real split, not a circular restatement. *(Grade: Fact — it is algebra. Its role is non-vacuity.)*

Enumeration validation of H(counts). The chain-rule `H(counts)` used in M1 was checked against brute-force enumeration of the full count simplex at n=50 (23,426 vectors) and n=100 (176,851 vectors): agreement to **1.4×10**^{1} and **6.4×10**^{1}, with the multinomial normalisation `Σ P(c)` recovering 1 to ~10^{1}. *(Grade: internal validation; no claim.)*

Capacity/blank harvest (C3, unbarred). Each cell has capacity ln4 but the source spends only H; the gap `ln4 - H = 0.173287 nats/symbol` is the redundancy a learned model can harvest — the same quantity as M3's knowledge discount, viewed from the cell rather than the eraser. Reported alongside, unbarred, per C3. *(Grade: Reading, and it is the same Reading as M3.ii.)*

Non-vacuity audit (per §5.3; performed before running)

Every committed bar is capable of failing under an incorrect implementation, and no check compares a quantity to itself or restates a definition:

- **M1** computes its three terms by three independent routes; a bug in any (a mis-enumerated simplex, a wrong multinomial coefficient, a normalisation error) breaks the identity. The specific trap — deriving a conservation term by subtraction — is avoided: `H(counts)` is a chain-rule computation, not `nH - E[ln C]`. The realized leg is a genuine Monte Carlo test.
- **M2** simulates the bounded-memory policy on the realized stream; a wrong charge or wrong discard rule shifts the toll off H. The independent cross-check (paid + retained = direct op-D(raw), to 7×10^{1}) confirms the split closes rather than assuming it.
- **M3** compares three realized tolls to three fixed closed forms. **Honest flag:** the *ignorant* toll (and M6's order-0 toll) is `-ln(¼) = ln4` for every symbol, so it is exact by construction, not statistical — it tests the op-D wiring, not a fluctuating mean. The genuinely statistical tolls are *learned* and *mismatched*, and those are where the discount and surcharge are measured.
- **M4** simulates archive-or-discard; the frontier and the `f·H` storage law both move under a miscoded storage/erasure account.
- **M5** is an exact computation whose slope fit fails if `E[ln C]` is wrong.

Notes for Alex (honest observations — none threaten a bar)

1. **The ignorant toll is deterministic, not measured.** Under the uniform protocol every symbol costs exactly `ln4`, so `M3-ignorant` and `M6-order0` land on ln4 to machine precision rather than "within 0.02." This is worth knowing so the two exact values aren't mistaken for suspiciously-good statistics — they are the capacity, by construction. The statistical teeth of M3 are the *learned* (1.21441) and *mismatched* (1.64699) tolls.
2. **M2's paid toll sits just above H, not below.** The exact ledger identity is `paid + retained = ingested`, and it closes to 7×10^{1} . Its relation to the *theoretical* floor H carries the stream's Monte Carlo fluctuation: here the ingested rate ran $+1.4\times10^{3}$ above H, more than the retained inventory (1.2×10^{3}) pulls down, so the paid toll is $+1.7\times10$ from H. Flagging so the *sign* of the deviation isn't a surprise. Reporting the retained inventory as an explicit storage line (your addition) is what makes this legible rather than mysterious.
3. **The M4 frontier is f-independent — that is the resolution in one number.** `storage + erasure = 1.21441` at every f. The dichotomy "balanced or stored" dissolves: it is always balanced *against the ingested information*, and "stored vs paid" is the free parameter f, not an alternative to conservation.
4. **The MDL slope is −1.5038, just off dead-centre.** Well inside ±0.05; the per-doubling decrements show a clean monotone approach to $-1.5 \cdot \ln 2$, so the small excess is the finite-n endpoint (n=50) transient in the log-linear fit. Noting it in case the n-range is ever changed.
5. **M6's gap exceeds MI by ~4×10^{3}** — Monte Carlo on the order-1 entropy-rate estimate; report-only and unbarred, as pre-registered.

Plain-language summary

A probe reads a stream, tidies its memory, and throws things away. The question was whether that tidying **balances** — every nat paid for — or whether the probe somehow ends up storing "net energy." The answer is a clean frontier.

- Everything the probe takes in has to go **somewhere**: either it is **paid off** (erased, at a Landauer cost) or it is **kept** (stored, still occupying memory). Add the two and you get back exactly what came in — every time, at every setting. There is no leak and no free lunch. We measured this directly: paid + stored = ingested, to sixteen decimal places.
- The **only** free choice is *how much to keep.* Keep nothing and you are a **summarizer**: you pay the full information rate and your memory stays flat. Keep everything and you are a **hoarder**: you pay nothing (reversible compression is free) and your memory grows at the full rate. Every mix in between sits on a straight line between those two.
- **Knowing the source is worth money.** An ignorant eraser pays the full cell capacity; an eraser that has learned the distribution pays less, and the saving is *exactly* the redundancy it learned. Knowledge is a discount on the price of forgetting.
- **Being confidently wrong is worse than knowing nothing.** An eraser using a wrong model pays a surcharge equal exactly to how wrong it is (the KL divergence) — and here that surcharge (0.43) is bigger than the whole redundancy (0.17), so the wrong model costs more than pure ignorance. The probe's model quality is readable straight off its own energy bill.
- **What survives forgetting is the model.** When the probe forgets the order of a block and keeps only the counts, the little that it keeps grows very slowly — like `1.5 · ln n`, the

textbook “model description length.” Forget the details and what is left, permanently, is the summary.

Everything landed where it was predicted to. Nothing had to be adjusted.

Mandatory honesty caveat (transplanted from the corpus’s energy-ledger discipline; §1)

In simulation, Landauer accounting is an **identity true by construction** — bijections are free and merges cost the entropy drop *because we compute the* ***`ln`*** *we predicted*, not because we measured a physical dissipation that could disagree. E2 therefore confirms that the probe’s ledger is a **consistent, correctly-structured accounting** and measures classical identities within it; it does **not**, and cannot, *risk* the framework’s carrier reading, which stays a **Lead pending physical measurement**. E2’s own contributions are the probe readings graded above.

Honesty boundary

The mathematics is classical and cited as such: Landauer’s principle and Bennett’s reversibility (the free/charged split), Shannon’s source-coding theorem and the mismatched-code identity (the floor H and the cross-entropy tolls ln4, H, H+KL), and Rissanen’s MDL (the `(k−1)/2 · ln n` residue). **E2 adds no mathematics to these.** What belongs to the sub-programme is the reading *on the probe’s own ledger*: the frontier resolution of “balanced or stored,” and the discount / surcharge / residue as measurements of a probe’s knowledge, error, and permanent residue against its own erasure bill. In simulation these are identities we computed rather than physical facts we risked, so nothing here is — or can become — evidence that the genesis system holds beyond the axioms that posit it; the carrier reading is not promoted. No leverage is claimed on any open problem. Boundary discipline inherited from the main spine, unamended.

The master document has not been edited — banking edits to the spine are performed by the lead instance on return.

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Lead verification addendum (appended on return; original text above untouched)

Independent rerun. The script was rerun by the lead, unmodified, in a separate environment: exit 0, **22/22 committed checks**, and the full run log **byte-identical to the submitted log** modulo the runtime PID line. The banking stands.

Code review of the non-circularity claims. The three M1 routes are genuinely independent *in code*, not merely in description: nH from the pinned closed form; E[ln C] as ln n! − Σ E[ln c !] via per-coordinate binomial sums; H(counts) via a chain rule whose conditionals are correctly derived (c |c ∼ Binom(n−c , p /(1−p)); c |c ,c collapsing to Binom(n−s, p /(p +p)) with s ∼ Binom(n, p +p)); plus brute simplex enumeration as validation. The anti-tautology clause of brief §5.3 is honoured in the implementation.

One relation noted for the record. M2's conservation cross-check (paid + retained = direct op-D(raw), 7×10^{1}) is algebraically entailed by A1's per-block route-equality plus the bookkeeping; as an implementation check it remains non-vacuous (two independently computed sums meeting), the same status the note itself assigns to A1. Recorded so the 10^{1} agreement is read as the split closing, not as a separate statistical result.

On the implementer's self-flags. The deterministic ignorant toll (and M6's order-0 toll) was identified by the implementing instance itself, in its own non-vacuity audit, before the lead's review — the correct call, correctly reasoned (it tests op-D wiring, not a fluctuating mean; M3's statistical teeth are the learned and mismatched tolls). Two experiments ago this class of observation had to come from the lead's addendum; the discipline is now being applied by implementers to their own work unprompted. Noted with approval.

Lead instance, July 2026.